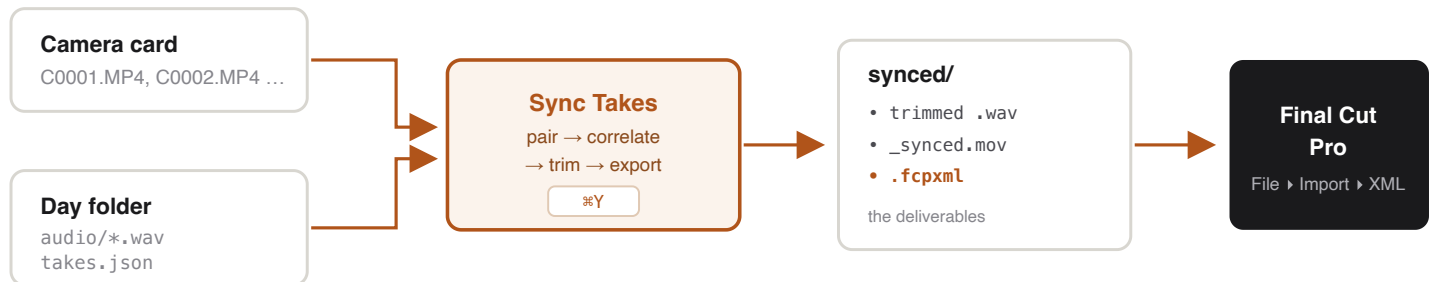


Getting synced takes into Final Cut Pro

CinemaHUD records a 24-bit WAV for every camera take and logs exactly when each one started. After the shoot, **Sync Takes** matches those WAVs to the clips off the card and writes an FCPXML that Final Cut opens as an event of ready-synced clips — camera audio muted, the good audio connected underneath.



The whole path. Nothing is copied into the day folder from the card — the FCPXML points at your clips wherever they live, and at the trimmed WAVs in synced/.

Before you start

- 1 **Copy the clips off the card onto a local disk.**
The FCPXML records each clip's path exactly as you dropped it. Drop them straight from the card and the event goes offline the moment you eject. Copy first, then drop the copies.
- 2 **Check the project frame rate in the HUD.**
Sync Takes inherits it and writes it into the XML as the format's frame duration. Changing it afterwards means exporting again.
- 3 **Know which day folder you want.**
Sync Takes opens on today. A shoot that ran past midnight, or one you are coming back to, needs *Choose Day Folder...* — pick the folder that contains `takes.json`.

RULE OF THUMB

The tighter the camera's own scratch audio, the better the match. Feeding the interface's line out into the camera's mic input makes correlation essentially bulletproof; the camera's built-in mic usually works too.

1 • Pair and sync

Open **Sync Takes** from the launch screen's **After the Shoot** panel, the Audio menu, or **⌘Y**. Drag the card's CLIP folder (or individual clips) onto the window — folders are walked for .mp4, .mov and .m4v.

Sync Takes

~/Movies/CinemaHUD/2026-09-16 **1** Choose Day Folder... Choose Clips... Sync All Export

Clip	Take	Duration	Offset	Confidence	Status
C0001	A_0001_C001	42.3 s	3.012 s	94 %	Synced
C0002	A_0001_C002	18.7 s	2.998 s	88 %	Synced
C0003	A_0001_C003	67.1 s	3.000 s	31 %	Low confidence (estimate used)
C0004	—	5.2 s	—	—	No take

Drop the card's CLIP folder anywhere on this window
Takes come from 2026-09-16 / takes.json

Exported to ~/Movies/CinemaHUD/2026-09-16/synced

2

3

4

1 — the day folder the takes are read from. **2** — the Take column is a picker; fix any mispairing by hand. **3** — *Sync All* cross-correlates, four clips at a time. **4** — *Export* (also **Return**) writes everything and reveals it in the Finder.

Reading the Status column

STATUS	WHAT IT MEANS — AND WHAT TO DO
Synced	Correlation found the offset at 50 % confidence or better. Nothing to do.
Low confidence	The match was weak, so the offset falls back to the estimate from the take log — usually within a frame or two, but check these in the editor.
Estimated	You picked the take by hand and haven't run <i>Sync All</i> yet. The offset is the take log's estimate.
No take	No take paired. Pick one in the Take column, or leave it — the clip still comes into Final Cut, just with camera audio only.
WAV missing	The take log names a WAV that isn't on disk. Check you're on the right day folder.
Failed	The message names the cause. Common one: a clip with no readable video track.

The Take picker lists only takes that finished cleanly. A take that was aborted, or that hit a write error, is deliberately not offered.

2 • What Export writes

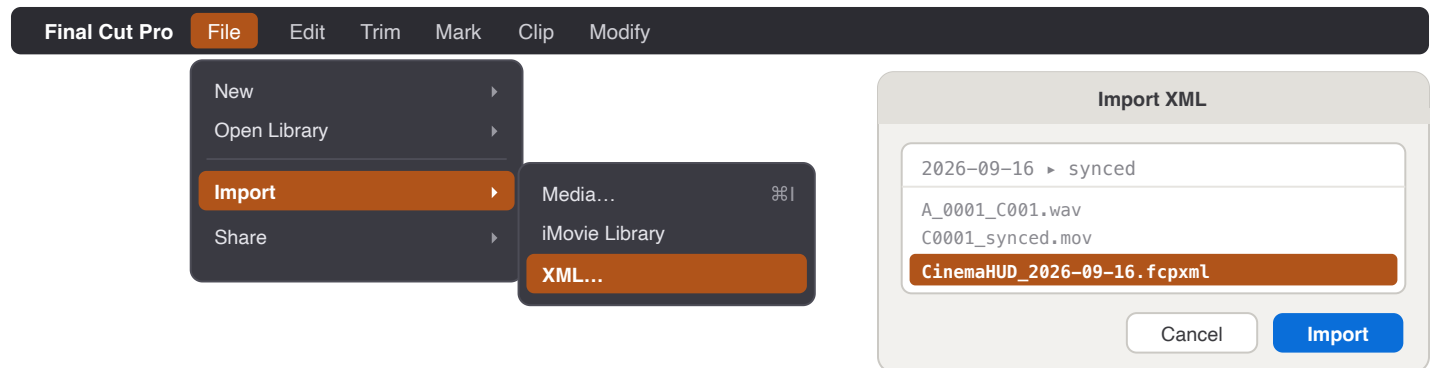
Everything lands in a synced folder inside the day folder. The Finder opens with the FCPXML selected.

```
~/Movies/CinemaHUD/2026-09-16/
├─ audio/                          the originals – never modified
│  └─ A_0001_C001.wav              48 kHz / 24-bit, 3 s pre-roll, BWF timecode
│  └─ A_0001_C002.wav
├─ takes.json                      the take log
└─ synced/                         ← everything Export writes
   └─ A_0001_C001.wav              trimmed to start exactly with the clip
   └─ A_0001_C002.wav
   └─ C0001_synced.mov             video + good audio + camera audio
   └─ C0002_synced.mov
   └─ CinemaHUD_2026-09-16.fcpxml ← the one you import
```

Two deliverables, one folder. Import the FCPXML for real sync clips you can still re-balance, or drag in the `_synced.mov` files for something self-contained.

3 • Import into Final Cut Pro

Open the library you want the footage in, then choose **File ▶ Import ▶ XML...** and pick `CinemaHUD_<day>.fcpxml`.

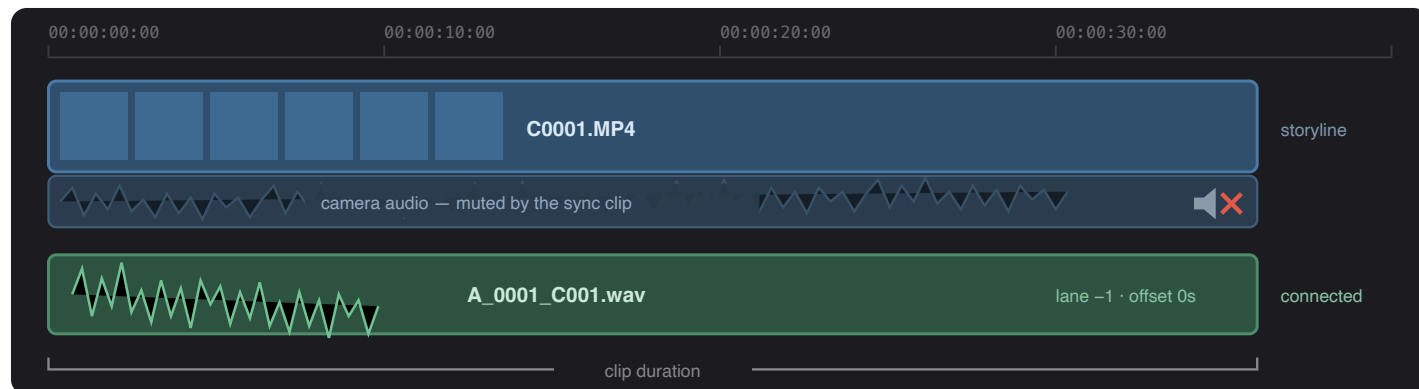


File ▶ Import ▶ XML... — *not Import Media*. Media would bring in the clips as loose footage with no sync at all.

Final Cut creates an event named **CinemaHUD <day>** holding one item per clip. If your library is set to copy media in, it will copy; leave it on “leave files in place” to keep referencing the originals where they sit.

4 · What you get, and how it's wired

Each paired clip arrives as a **sync clip**, already assembled. Open one and the anatomy is simple: the camera clip sits on the storyline with its own audio switched off, and the trimmed WAV hangs beneath it on lane -1, starting at zero.



The sync clip, opened up. The WAV was trimmed on export to begin exactly at the clip's first frame, so its offset is a flat zero — there is no slipping left to do. Both tracks carry the **dialogue** role, so a role-based mix treats them as one.

Working with it

- **Just cut.** Drag sync clips to the timeline like any other clip; the good audio travels with them.
- **Need the camera audio back?** Open the sync clip, select the storyline clip and turn its audio back on. It's muted, not removed — useful as a reference or a safety.
- **Multi-channel takes** come in with their channel count and sample rate as recorded, and the track names you set are in the file's metadata.
- **Timecode.** Every WAV carries Broadcast Wave time-of-day timecode matching the HUD's readout, shifted to account for the trim, so it still lines up if you'd rather sync by timecode yourself.
- **Unpaired clips** come in as ordinary clips with camera audio only — useful to spot what never got a take.

IF YOU'D RATHER SKIP THE XML

Import the `_synced.mov` files instead. Each is the original video passed through untouched, with the interface audio as track 1 and the camera audio as track 2 — one self-contained file per clip, nothing to relink. The trade is that you can no longer treat the two audio sources as separate sync sources inside Final Cut.

If something looks wrong

SYMPTOM	CAUSE AND FIX
Clips import offline / “missing media”	The XML points at the path you dropped from. If that was the card, reconnect it — or better, copy the clips to a local disk, run Sync Takes again against the copies, and re-import.
“Invalid edit with no respective media”	An FCPXML exported before the duration fix. Export again from a current build — the WAVs on disk are fine, only the XML was wrong.
Final Cut rejects the file outright	The document declares FCPXML 1.11. A much older Final Cut won't read it; update, or use the <code>_synced.mov</code> route instead.
A clip is a frame or two out	It was a Low confidence row, so the offset came from the take log rather than the waveform. Nudge it in the editor, or re-shoot the scratch-audio path for the next day.
Audio is a whole take out	Mispaired. Pairing is by order and length, so a clip deleted on the camera can shift everything after it. Fix the Take column by hand, run <i>Sync All</i> , export again.
Frame rate looks wrong in Final Cut	The XML's frame rate comes from the HUD's project frame rate at export time. Set it, then export again.
A <code>_synced.mov</code> has one audio track, not two	The offset landed past the end of the WAV — the take and the clip don't actually overlap. That row would have shown low confidence; re-pair it.
Takes list is empty	Wrong day folder, or the log was unreadable and got moved aside to <code>takes.json.corrupt-...</code> — the app says so when that happens.

Quick reference

Open Sync Takes	The launch screen's After the Shoot panel, the Audio menu, or <code>⌘Y</code>
Export	<code>Return</code> in the Sync Takes window
Recordings	<code>~/Movies/CinemaHUD/<day>/audio/</code>
Deliverables	<code>~/Movies/CinemaHUD/<day>/synced/</code>
Import as	File ▶ Import ▶ XML... — <i>not</i> Import Media
Event name	CinemaHUD <day>
Confidence floor	50 % — below it, the take log's estimate is used
Pre-roll / post-roll	3 s before the REC press, 1 s after the body stops

STATUS

The first real Final Cut import failed with “*Invalid edit with no respective media*” on the connected WAV: durations were rounded to the nearest frame, so a clip that didn’t land on a frame boundary claimed up to half a frame more audio than the trimmed file holds. Durations bounded by real media now round down, and the WAV states its length in samples rather than frames. Re-export before importing — an FCPXML written before that fix still carries the bad durations.